

A Survey of Emerging Topologies

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The rise of AI, especially in the form of large language models like ChatGPT,

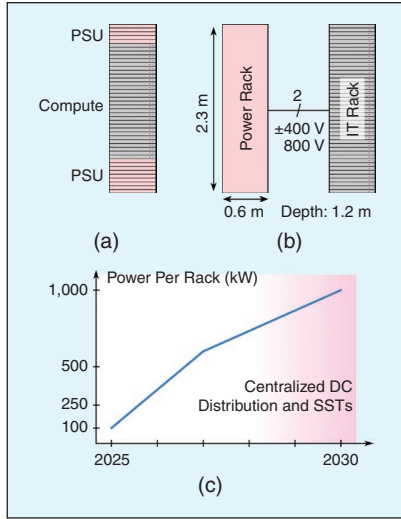


FIGURE 1 – (a) Conventional server rack containing compute servers as well as PSUs. (b) Disaggregated rack concept [1], [4] with dedicated power and IT racks, enabling the (c) projected increase in IT rack power consumption over the coming five years based on [5].

drives a transition from CPU servers to GPU servers to support massive parallelization of operations during AI model training. A single GPU chip will draw up to 10 kA of current [1], and thermal design powers will reach 15 kW and beyond in the coming years [2]. The correspondingly increasing power consumption density of AI servers cannot be supported anymore by the conventional server rack configurations that intersperse power supply units (PSUs) and compute servers [see Figure 1(a)] [3]. Instead, the “disaggregated rack” concept [1], [4] moves the power equipment into a dedicated power rack that supplies power to the compute servers in the adjacent IT rack at 800 VDC [see Figure 1(b)]. This approach supports IT rack power levels that will increase into the 1-MW range over the next few years [see Figure 1(c)]. Typical power rack dimensions (e.g., Open Compute Diablo 400 [4]) result in a power density of about 0.65 MW/m³, a value that we will revisit later.

At the data center level, the power rack concept still relies on LVAC distribution at typical voltage levels of 415 V or 480 V line-to-line root mean square (RMS) [see Figure 2(a)]; it is thus a temporary solution (a white space retrofit) that enables an increase of the

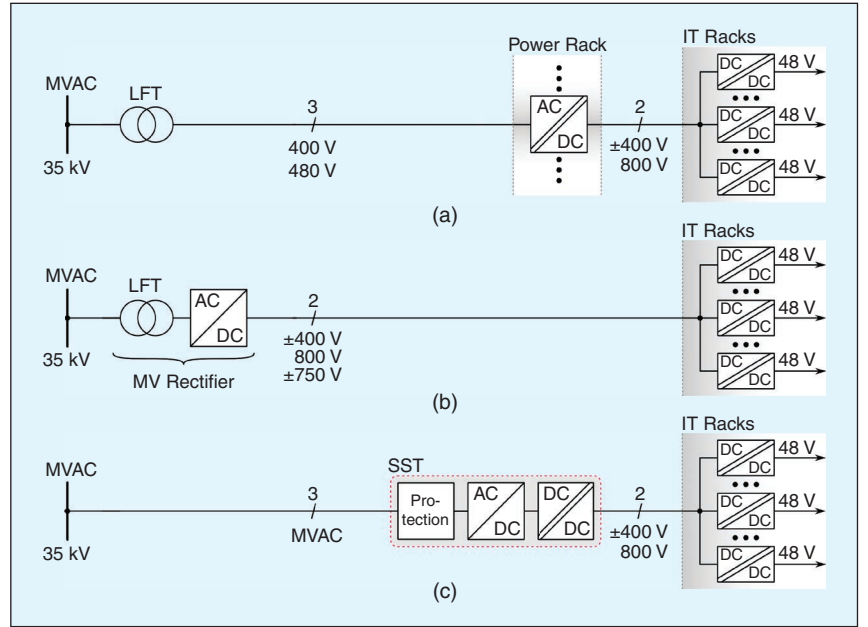


FIGURE 2 – Data center power distribution architectures facilitating 1 MW per rack. (a) LVAC distribution with sidcar power racks. (b) Central MVAC–LVDC conversion and LVDC distribution. (c) Future MVAC distribution and replacement of the power rack from (a) by an MVAC–LVDC SST, which thus should feature a power density approaching 1 MW/m³. The power distribution busbars, power racks, and/or SSTs could be placed next to, above, or below the IT racks. *Note:* Protection devices, etc., are not shown for clarity. LFT: low-frequency transformer.

compute density and gaining experience with 800-VDC systems on a local scale [6].

However, using LVDC instead of LVAC for the power distribution inside of the data center improves the losses in the distribution busbars *and* reduces the amount of conductor material required, as visualized in Figure 3(a) and (b). With

$$P_{\text{loss,AC}} = 3 \left(\frac{P/3}{V_{\text{AC}}} \right)^2 R$$

$$\text{and } P_{\text{loss,DC}} = 2 \left(\frac{P}{2V_{\text{DC}}} \right)^2 R$$

we have $P_{\text{loss,DC}}/P_{\text{loss,AC}} = 0.75$ if the LVDC voltage is selected as $V_{\text{DC}} = \sqrt{2} V_{\text{AC}}$ with V_{AC} denoting the AC line-to-neutral RMS voltage, i.e., assuming roughly the same isolation stress. Further, the LVAC distribution system requires three conductors with resistance R , whereas the LVDC distribution requires only two; i.e., it reduces the material consumption by 33%.

Therefore, road maps like [6] expect a next development step toward LVDC distribution (800 VDC or, depending on the grounding scheme, ± 400 VDC) with central facility-level MVAC–LVDC conversion [see Figure 2(b)] [1], [3],

[6], [7], [8], [9], whereby a typical MVAC input voltage level is 35 kV (line-to-line RMS). LVDC distribution comes with challenges of its own, especially regarding protection because of the absence of a current zero crossing; therefore, solid-state circuit breakers (SSCBs) might play an important role [6]. Initially, the MV rectifiers could be based on low-frequency transformers (LFTs) and downstream AC–DC converters or high-power 12-pulse rectifiers, as currently employed, e.g., for

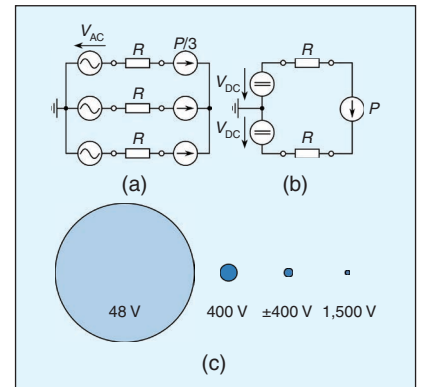


FIGURE 3 – (a) Three-phase AC distribution and (b) DC distribution using $V_{\text{DC}} = \sqrt{2} V_{\text{AC}}$. (c) Conductor cross sections for equal distribution losses for a given power transfer at different DC voltages.

electrolyzers; we will discuss such options briefly in the section “LFT-Based Short-Term Solutions.” For example, [6] suggests a reference LVDC data center architecture based on 17.5-MW power domains fed by five 3.3-MW MV rectifiers (four-out-of-five redundancy).

A further reduction of the conductor material requires an increase in the (DC) voltage level, as indicated in Figure 3, because for given losses, the conductor cross-sectional area scales with $1/V_{DC}^2$. Today, 800 VDC (or ± 400 VDC) are considered for data centers, which might increase in the future to 1,500 VDC (typically

realized as ± 750 VDC) [8], the current standard in large-scale photovoltaic (PV) installations [10]. Recently presented server power supplies with 800-VDC input and isolated 12-VDC output are typically modular and hence scalable to higher DC input voltage levels [11], [12].

Ultimately, SSTs could enable compact MVAC–LVDC conversion close to the server racks and hence MVAC power distribution *inside* of the data center [see Figure 2(c)]. Based on [13] and [14], this article first identifies a power density barrier of today’s state-of-the-art industrial MVAC–LVDC SST demonstrators (in the section “State-of-the-Art SST Topologies and Limitations”), which employ the fully modular two-stage topologies originally investigated for (single-phase) traction applications. Then, the section “Next-Generation SST Topologies” introduces emerging and future (quasi) single-stage three-phase SST topologies that have the potential to achieve a power density target of 1 MW/m³, motivated by the 0.65 MW/m³ of the LV power racks discussed previously. To complete the picture, the section “LFT-Based

Short-Term Solutions” discusses LFT-based MVAC–LVDC conversion solutions that are suitable for intermediate steps using LVDC distribution. Finally, the section “Conclusion and Outlook” concludes the article with an overview of future research vectors and an outlook on future MVDC distribution employing MVDC–LVDC SSTs near the server racks.

State-of-the-Art SST Topologies and Limitations

The main motivation for SSTs follows from the basic scaling law of a transformer, as shown in Figure 4, which directly relates an increase in operating frequency f to a significant reduction in construction volume:

$$A_w A_c \propto \frac{S}{f},$$

$$\text{and hence } L^4 \propto \frac{1}{f}$$

$$\text{and } L^3 \propto \frac{1}{f^{\frac{1}{3}}}$$

without considering isolation requirements and other practical aspects. A_w and A_c are the winding window area and the core cross-sectional area,

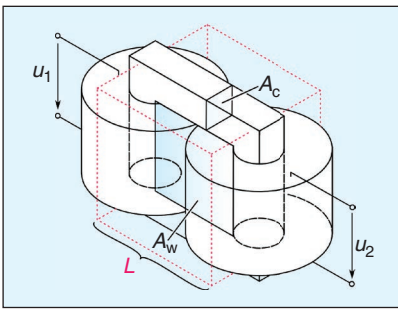


FIGURE 4 – Conceptual representation of a transformer with winding window area A_w , core cross-sectional area A_c , and characteristic length L .

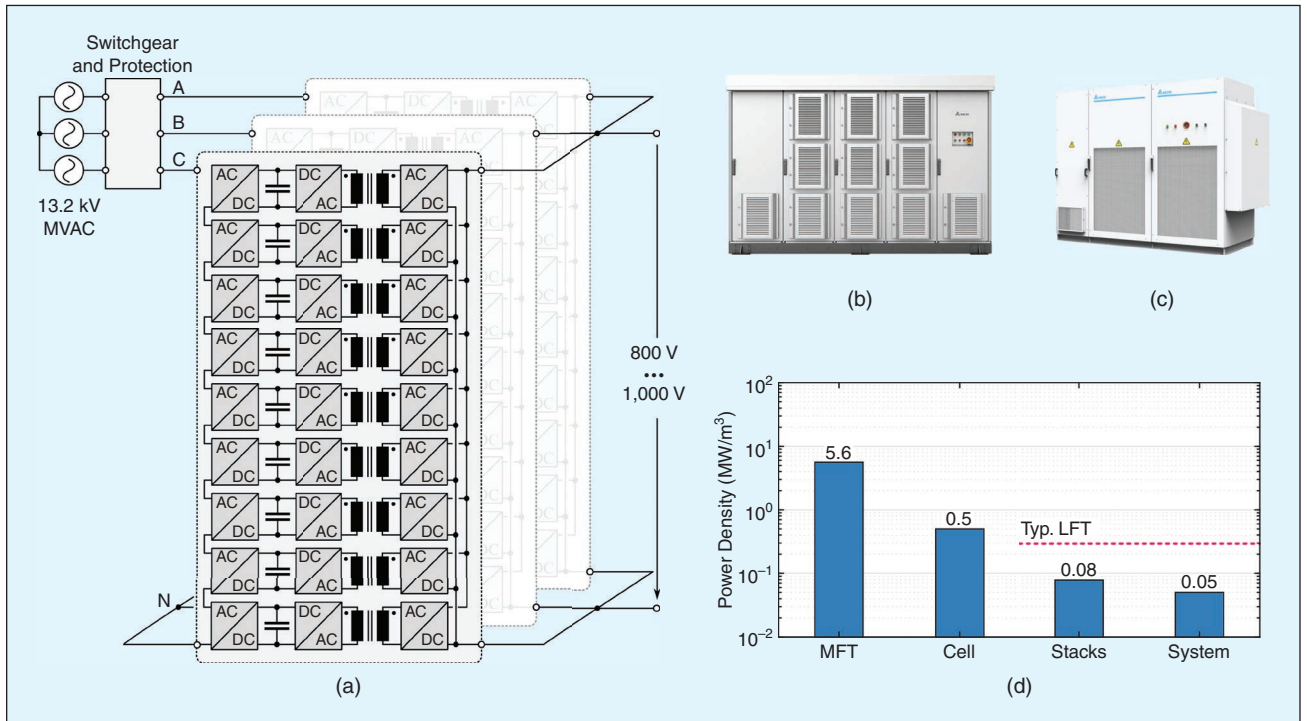


FIGURE 5 – Industrial two-stage SST system example. (a) Topology and realization of (b) first-generation (400-kW) [15] and (c) second-generation (1,200-kW) prototypes in a similarly sized cabinet [16]; note the threefold increase in power density. (d) Power densities of the first-generation prototype’s subassemblies, indicating the “modularization penalty” (see Figure 6) and the impact of the per-cell energy storage requirements to buffer the twice-mains-frequency power pulsation in each phase string.

respectively, S is the nominal VA rating, and L a characteristic length, as illustrated in Figure 4. Hence, the power density improves with the operating frequency as

$$\frac{P}{L^3} \propto L \cdot f.$$

Thus, replacing an LFT with a power electronic converter that operates a transformer at an elevated frequency, i.e., that uses a medium-frequency transformer (MFT), is a direct path toward compact MVAC–LVDC conversion, which was initially explored for traction applications, e.g., in [17], [18], [19], and [20]. As mentioned previously, today’s state-of-the-art SSTs targeting high-power electric vehicle (EV) charging or data center applications typically employ a modular two-stage approach [see Figure 5(a)]: Each MVAC phase connects to a string of identical converter modules consisting of a dedicated single-phase AC–DC rectifier stage, a local DC buffering capacitor, and a downstream isolated DC–DC converter containing an MFT. These modules are connected in an input-series-output-parallel (ISOP) configuration to handle the MV input. Figure 5(b) and (c) show recently published industrial megawatt-class SST demonstrator systems, indicating a threefold improvement in power density between first- and second-generation systems—indicating that, with two-stage SSTs, significant future

improvements in power density are also possible through optimized engineering, e.g., regarding isolation coordination and/or MFT design. However, closer inspection of the power densities of the first-generation SST subassemblies [see Figure 5(d)] reveals that the very high compactness of an MFT does *not* ultimately translate into high system-level power densities.

The first reason lies in the topology’s phase-modular nature, which requires buffering of (at least part of) the characteristic twice-mains-frequency power pulsation on the MV side of each converter cell using large film capacitors. The second reason is the “modularization penalty” visualized in Figure 6: each cell contains an isolation barrier that must be rated according to the full MVAC voltage processed by the SST. This overhead penalizes designs with too many cells, i.e., those employing too low per-cell DC voltages (which, however, are limited by available power semiconductors) and/or too low a power rating per cell. The latter implies an MVAC-voltage-dependent lower boundary for the reasonable power rating of an SST.

As a result, today’s industrial SST demonstrator systems do not significantly outperform LFT-based solutions and are limited to power densities of about 0.2 MW/m³ at efficiencies of

about 98.5% (see Figure 7). To replace today’s LVAC–LVDC power racks with MVAC–LVDC SSTs supporting MVAC distribution up to the servers, a step across the power density barrier is required. Next-generation MVAC–LVDC SSTs should thus target a power density of 1 MW/m³ (of the overall SST system) at an efficiency of 99%.

Next-Generation SST Topologies

In the following, based on the classification of MVAC–LVDC conversion solutions shown in Figure 8, we discuss next-generation SST topologies that have the potential to meet these ambitious targets. Single-stage topologies concentrate the AC–DC rectifier and the isolated DC–DC stage in a single, isolated AC–DC converter and therefore dispose of the potentially large capacitors otherwise needed for buffering the single-phase twice-mains-frequency power pulsation (note that the total LVDC output power is the sum of power drawn from all three MVAC phases and hence constant). Quasi-single-stage topologies aim for maximal simplification, concentration, or integration of the rectifier stage, e.g., using three-phase diode rectifiers. SSTs are often realized from modules in an ISOP configuration and/or feature a phase-modular structure; i.e., they are at least partly modular.

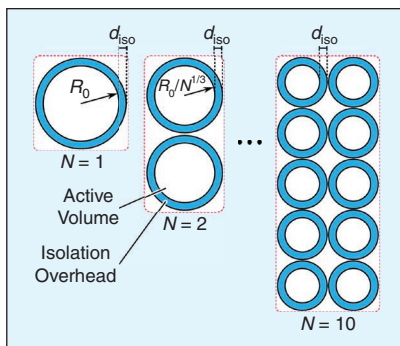


FIGURE 6 – Simplified intuition for the “modularization penalty” [14]: spheres of constant power density (representing, e.g., the active parts of a converter module) and with a constant isolation sheet around the active volume. Clearly, splitting the total power into many modules increases the overall size/ reduces the power density.

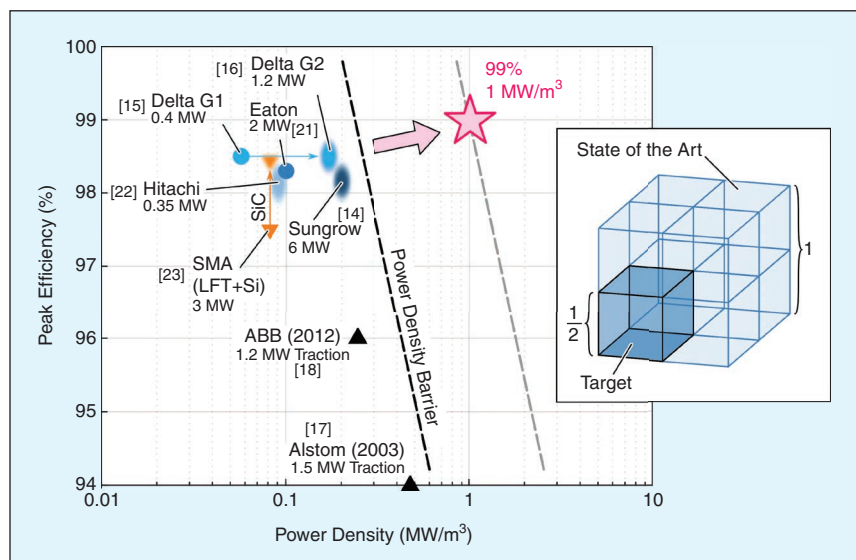


FIGURE 7 – Overview of state-of-the-art MVAC–LVDC conversion solutions, especially those including the full-scale industrial SST demonstrator systems reported in the recent literature. Achieving the power density target of 1 MW/m³ requires a size reduction by a factor of 5–10. (Updated version of a figure first shown in [13].)

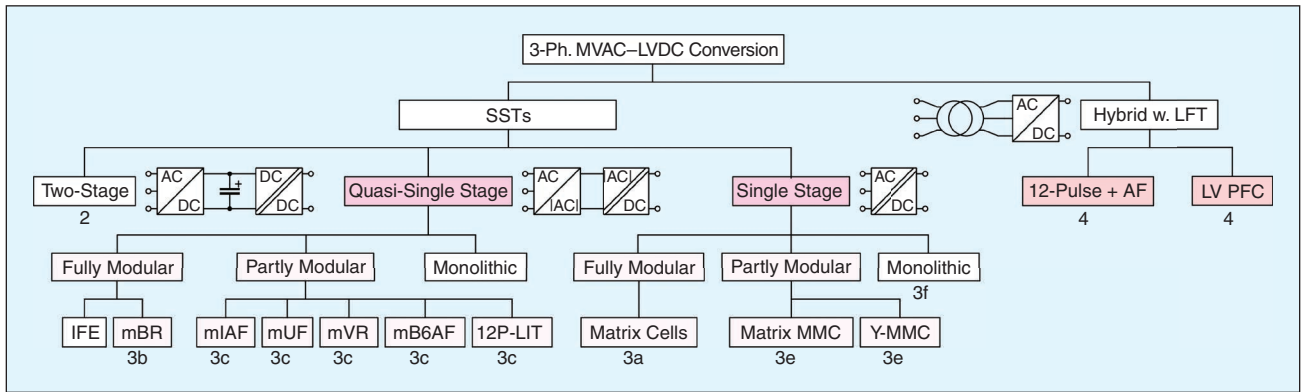


FIGURE 8 – Classification of (three-phase) MVAC-LVDC conversion systems with selected examples. (Highlighted examples are discussed in sections of this article: 2, “State-of-the-Art SST Topologies and Limitations”; 3a, “Single-Stage Fully Modular SSTs”; 3b, “Quasi-Single-Stage Fully Modular SSTs”; 3c, “Quasi-Single-Stage Partly Modular SSTs”; 3e, “Single-Stage Partly Modular SSTs”; 3f, “Single-Stage Monolithic SSTs”; 4, “LFT-Based Short-Term Solutions.”) AF: active filter; mBR: modularized bridge rectifier; mIAF: modular integrated active filter; mUF: modular unfolded; mVR: modular Vienna rectifier; MMC: modular multilevel converter. IFE: isolated front-end; mB6AF: modular B6 + active filter; 12P-LIT: 12-pulse rectifier with line-interface transformer.

Topologies combining both degrees of modularity, i.e., those that are phase modular *and* show a modular structure in each phase, are fully modular. The following examples clarify this classification further.

Single-Stage Fully Modular SSTs

Starting from the state-of-the-art fully modular SST demonstrator discussed previously and shown in Figure 5(a), which features two-stage converter cells, it is immediately clear that these converter cells could be replaced by single-stage isolated AC-DC converter cells while retaining the phase-modular ISOP configuration of the cells (see Figure 9). Such concepts have been considered for insulated-gate bipolar transistor (IGBT)-based early traction SSTs [19], [24], and

more recently also using SiC-based AC-DC dual-active-bridge (DAB)-type modules [25].

The topologies are still phase modular, and, for a given power semiconductor blocking voltage and utilization thereof, the total number of converter cells N is proportional to the phase voltage amplitude [as for the two-stage system in Figure 5(a)]. In contrast to two-stage converter cells, however, the MFTs now process pulsating power, i.e., there is no MV-side energy storage required, and more compact realizations should be achievable (despite the higher rated power of the MFTs); the single-stage power conversion should also result in a higher conversion efficiency. However, the modularity penalty still applies; i.e., sufficiently high power ratings per module are

important, and the future availability of SiC power transistors with blocking voltage ratings of several kilovolts will be beneficial.¹

Quasi-Single-Stage Fully Modular SSTs

Instead of single-stage conversion cells directly providing isolated AC-DC conversion, an alternative quasi-single-stage approach employs a full-wave diode rectifier to provide a rectified AC voltage that is strictly positive to a downstream isolated DC-DC converter [29] that advantageously does not need AC switches and/or intricate commutations sequences. Again, the DC-DC converters process pulsating power, and there are no energy storage capacitors on the MV side.

Taking this idea one step further, the modularized bridge rectifier (mBR) MVAC-LVDC SST topology shown in Figure 10 reduces the single-phase unfolders to a single diode per converter cell, again ensuring strictly positive input voltages for the isolated DC-DC converters [30], [31], [32]. Such high-power isolated DC-DC converter modules using SiC technology are available (e.g., a series-resonant converter with 300 kW in [33] or a 500-kW DAB in [34]).

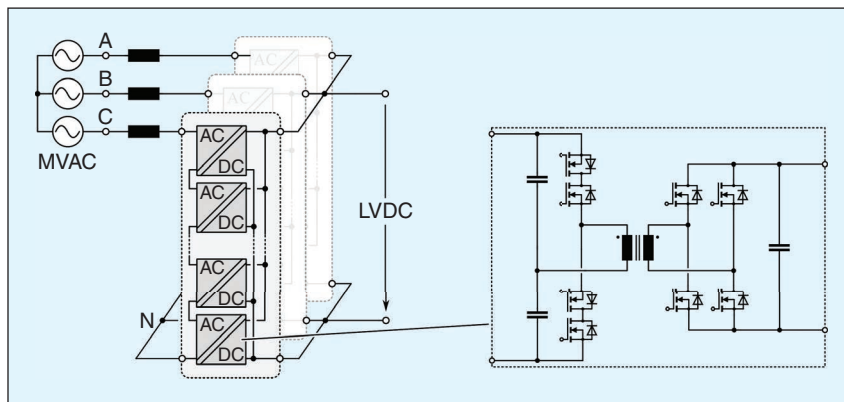


FIGURE 9 – Single-stage fully modular SST consisting of phase stacks with single-stage isolated AC-DC converter modules [19], [24], [25]. Note that the AC-side BDSs of the AC-DC converter module could alternatively be realized with monolithic 650-V GaN BDSs [27] or, offering higher blocking voltages, with SiC BDSs realized from the tight integration of two SiC MOSFETs on a common substrate [28] and/or in a common package. BDS: bidirectional switch.

¹Note that variations of this approach employ open-loop isolated AC-|AC| converter cells but require an LV-side |AC|-DC converter for power flow and output voltage regulation [26]; i.e., they are essentially two-stage systems with, however, simplified MV-LV conversion stage.

Advantageously, the mBR features a very high degree of modularity as only a single type of DC–DC module is used, achieves full power-factor-correcting (PFC) operation at the mains, and inherently supports multiport operation with an MVDC port between terminals P and N [32]. On the other hand, the utilization of the modules is relatively low, resulting in a high total module count (N is about three times higher than for the state-of-the-art two-stage SST from Figure 5) [35] and hence a high sensitivity to the modularity penalty.

Quasi-Single-Stage Partly Modular SSTs

Giving up the full phase modularity but keeping the idea of using only isolated DC–DC converter modules leads to concepts based on three-phase diode rectifiers (three-phase unfolders). Figure 11(a) shows a highly robust three-phase front end with a line interface transformer (LIT) and a 12-pulse rectifier to achieve a certain degree of PFC [36]—possibly sufficient depending on applicable grid codes. Advantageously, the power rating of the LF magnetic component, the LIT, is only about 15%–20% [37]. A single ISOP string of isolated DC–DC converters is needed [38], which further operates from a reasonably constant DC voltage with a maximum value given by the MVAC line-to-line voltage amplitude. Hence, only relatively few converter modules (about half as many as

for the state-of-the-art two-stage SST from Figure 5), which are further only DC–DC converters, are needed [35]. (The section “Remark: Realization of MVdc-LVdc Conversion” discusses alternative realization options for MVDC–LVDC conversion.)

Similarly, the topology shown in Figure 11(b) uses a simple B6 diode rectifier to generate a six-pulse-shaped quasi-DC voltage between the nodes P and N, where again an ISOP string of isolated DC–DC converters controls a constant power flow to the LVDC side. In addition, an active filter (AF) rated at 25% of the nominal power [35] ensures sinusoidal mains currents. To handle the MVAC level, the AF uses a phase-modular structure with series-connected full-bridge AC–DC converter cells, and the total AF module count almost (80%) equals that of the state-of-the-art two-stage SST from Figure 5. The system shows a higher complexity and realization effort (the AF converter cells require energy buffering capacitors) than the LIT-based variant discussed previously, but it also achieves better performance (sinusoidal mains currents).

The modular Vienna rectifier (mVR) topology [35] from Figure 12(a) bears close similarity to the B6 bridge with the AF [see Figure 11(b)], but it features a connection between the AF star point and the midpoint of the ISOP DC–DC converter string, enabling an advantageous 2/3-PWM (pulse-width modulation) operation

of the filter branches [i.e., only two out of the three AF phase strings operate with high-frequency (HF) switching at any given time] and hence lower switching losses with similar realization effort.

Integrating the AF by moving it to the DC side, resulting in an integrated active filter (IAF) [41], further simplifies the realization of the modular IAF (mIAF) SST shown in Figure 12(b) [39], [40]. Whereas the DC-side AF consists of about as many converter cells as in the state-of-the-art two-stage SST from Figure 5, the cells contain only a single half bridge and do not require significant energy storage. On the other hand, the IAF relies on low-frequency-operated

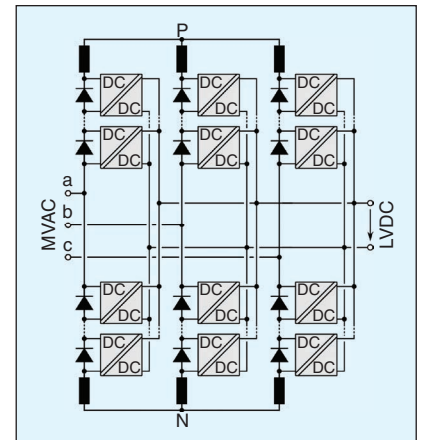


FIGURE 10 – Fully modular quasi-single-stage SST known as the mBR [30], [31], [32]. The diodes in parallel to the inputs of the DC–DC converters ensure strictly positive voltages, and the DC–DC converter control ensures PFC operation. PFC: power-factor correcting.

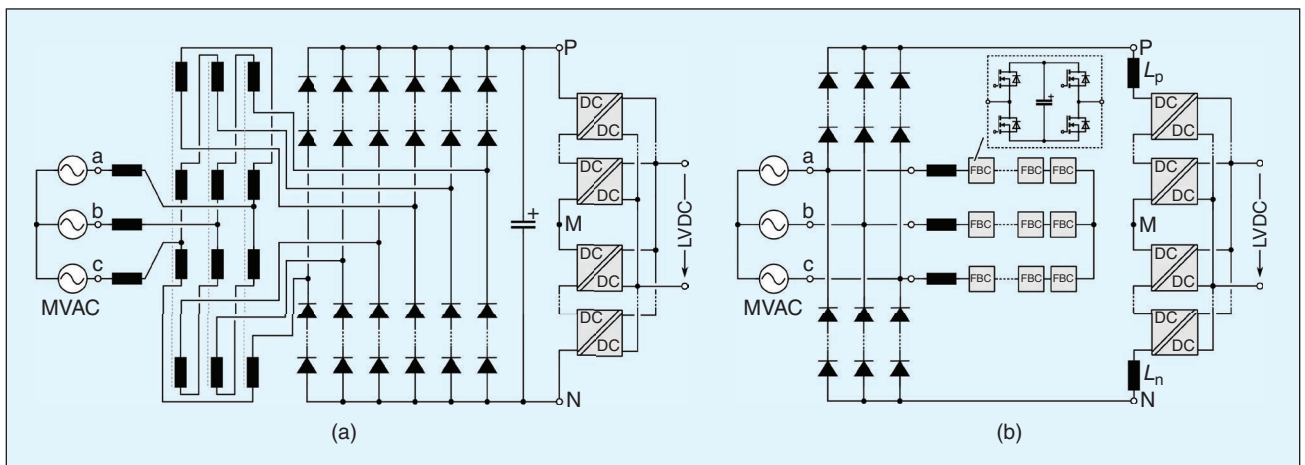


FIGURE 11 – Partly modular quasi-single-stage SST topologies based on passive three-phase unfold stages: (a) a 12-pulse rectifier with LIT front end [36], [38] or (b) a B6 diode bridge and an AF for achieving sinusoidal mains currents. LIT: line-interface transformer. FBC: full-bridge cell.

phase-selector switches (PSSs) to inject the filter current into the mains phase that is not connected to P or N via the passive diode rectifier at a given time instant. Note that the direct series connection of power transistors for realizing the PSSs requires means to ensure static and dynamic voltage balancing.

The ISOP DC–DC converter string can take over the functionality of the AF by connecting the string midpoint directly to one of the phase terminals using the PSSs [see Figure 12(c.i)]. This concept has been proposed for LV (i.e., with two DC–DC converters instead of two strings of DC–DC converters) in [42], and it was used in a similar form for an MVAC-connected STATCOM application using thyristors instead of diodes [43]. Advantageously, this modular (three-phase) unfold (mUF) rectifier requires only a single type of converter module, and there is no

need for energy storage on the MV side. On the other hand, the power and voltage processed by each of the string halves goes down to zero several times in each mains period [see Figure 12(c.ii)], which could be supported, for example, by operating only a few of the modules during times of low input voltage [44]. Still, each of the two DC–DC converter string halves must be rated for (almost) the full line-to-line voltage amplitude, resulting in almost twice as many DC–DC converter modules compared to the other topologies described in this section.

Note that all topologies support, in principle, bidirectional power flow if the rectifier diodes are replaced by switches (e.g., IGBTs or integrated gate-commutated thyristors). Interested readers can find a more detailed initial comparative evaluation of these (and further) three-phase-unfold-based SST topologies in [35].

Remark: Realization of MVDC–LVDC Conversion

The previously discussed SST topologies require high-power isolated MVDC–LVDC converters, which until now have been shown as ISOP arrangements of several isolated DC–DC converter modules [see Figure 13(a)]. Even though typically only about 50% of the modules compared to the state-of-the-art phase-modular two-stage solutions are needed, still each module contains an MFT with full isolation voltage rating; i.e., the modularization penalty still applies.

Instead, the isolated MVDC–LVDC conversion could be realized using a single quasi-two-level-(Q2L)-operated modular multilevel converter (MMC) on the MV side and only a single, fully rated MFT [see Figure 13(b)] [45], [46], [47]. As each half of the MMC bridge leg must support the full MVDC voltage, the topology requires twice the number of modules compared to the

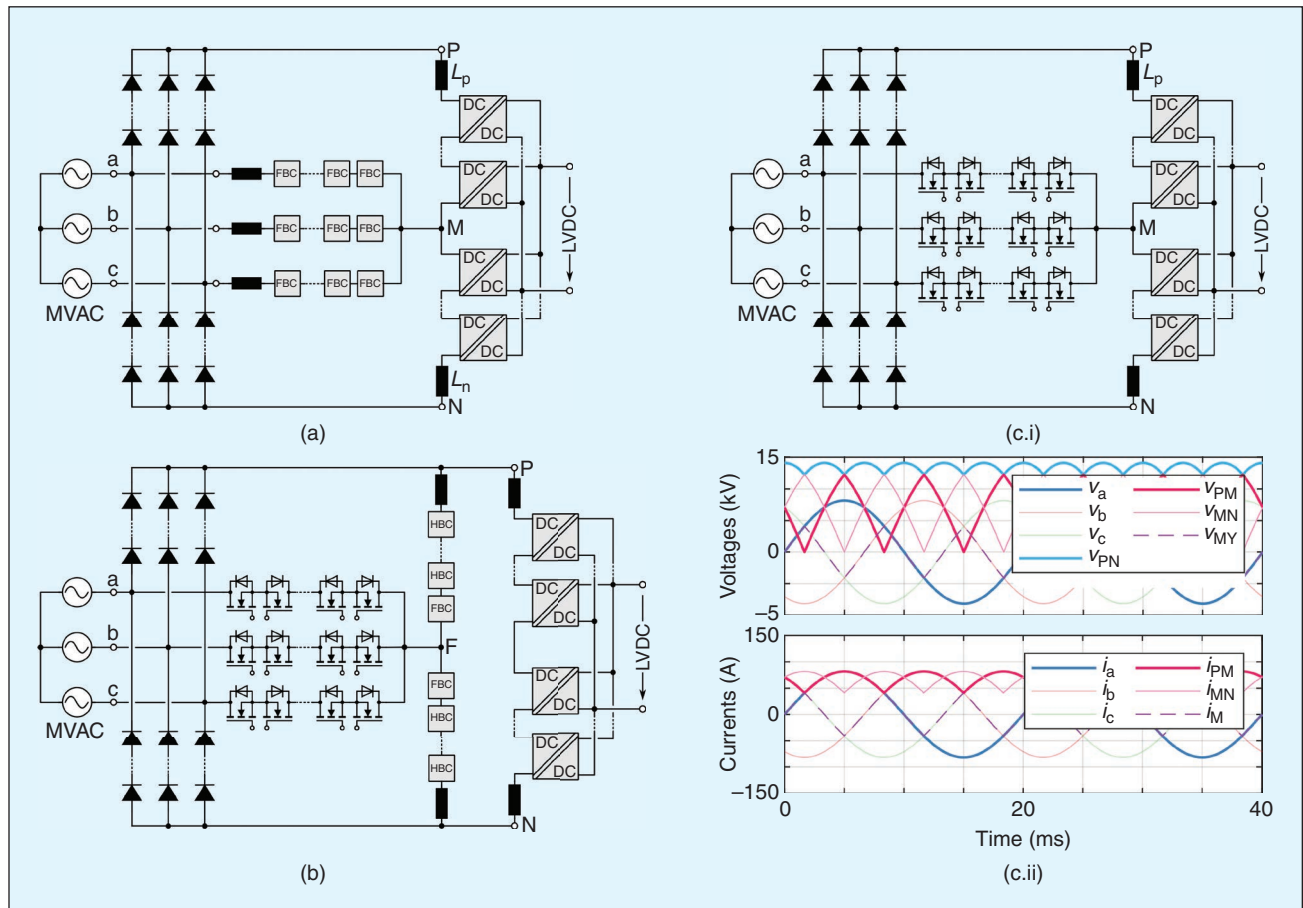


FIGURE 12 – Partly modular quasi-single-stage SST topologies based on three-phase unfolders. (a) mVR SST [35], (b) mIAF SST [39] [40], and (c.i) mUF SST with (c.ii) key waveforms (based on [35]). HBC: half-bridge cell.

ISOP variant from Figure 13(a)—but each module consists only of a single half bridge and a (small, from the Q2L operation) capacitor and does *not* contain an MFT.

Single-Stage Partly Modular SSTs

Driven by the same motivation to avoid the modularization penalty by reducing the number of isolation transformers—ideally to just one—a similar approach as just discussed for MVDC–LVDC conversion results in the single-stage partly modular MVAC–LVDC SST topologies shown in Figure 14. The first variant features an MMC front end using full-bridge cells, which performs direct three-phase AC to single-phase AC conversion [see Figure 14(a)] [48], [49]. Being an MMC, the installed power in terms of converter modules is relatively high; i.e., the clear advantage of a single MFT is paid for by requiring almost as many (80%) full-bridge cells per arm as the state-of-the-art two-stage SST from Figure 5 in total [3], and the overall number of cells is almost five times higher.

Instead, a phase-modular MMC front end using Q2L modulation to supply a three-phase MFT [50] requires fewer cells per arm (sufficient to block twice

the phase voltage amplitude only), and hence the total number of cells is only twice that of the state-of-the-art two-stage SST from Figure 5. The cells are further only half-bridge cells; an alternative realization based on full-bridge cells—essentially mimicking AC switches as used in LV versions [51], [52]—would halve the cell count.

Single-Stage Monolithic SSTs

Driven by the recent availability of LV monolithic bidirectional GaN power transistors, single-stage isolated three-phase LVAC–LVDC topologies have received renewed attention, i.e., topologies based on a direct three-to-single-phase matrix converter [see Figure 15(a)] [53], [54] or, alternatively,

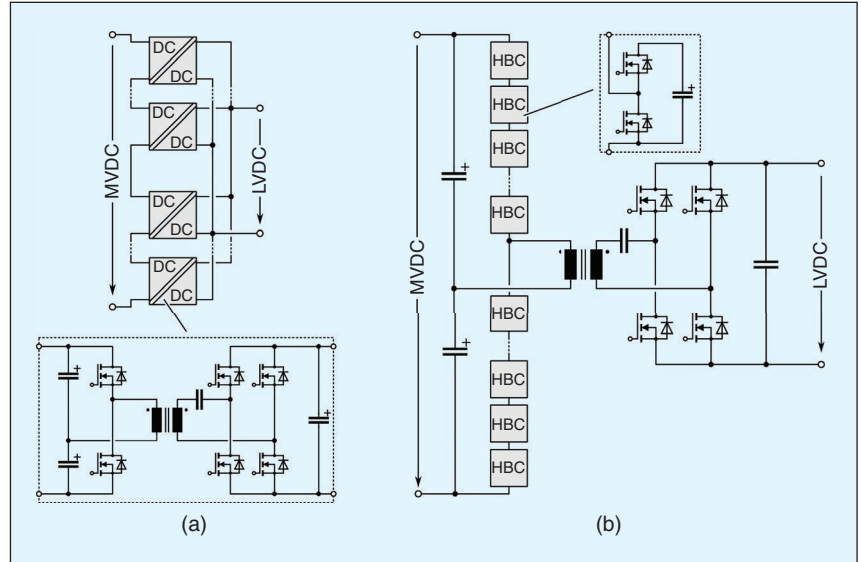


FIGURE 13 – Realization options of MVDC–LVDC conversion. (a) Fully modular ISOP configuration of individual DC–DC converter stages. (b) Partially modular variant with an MMC MV-side switching stage operating in Q2L mode and a single MFT [45], [46], [47]. The secondary side winding could be realized from several parallel windings with dedicated rectifiers to handle the high current load. Q2L: quasi-two-level mode.

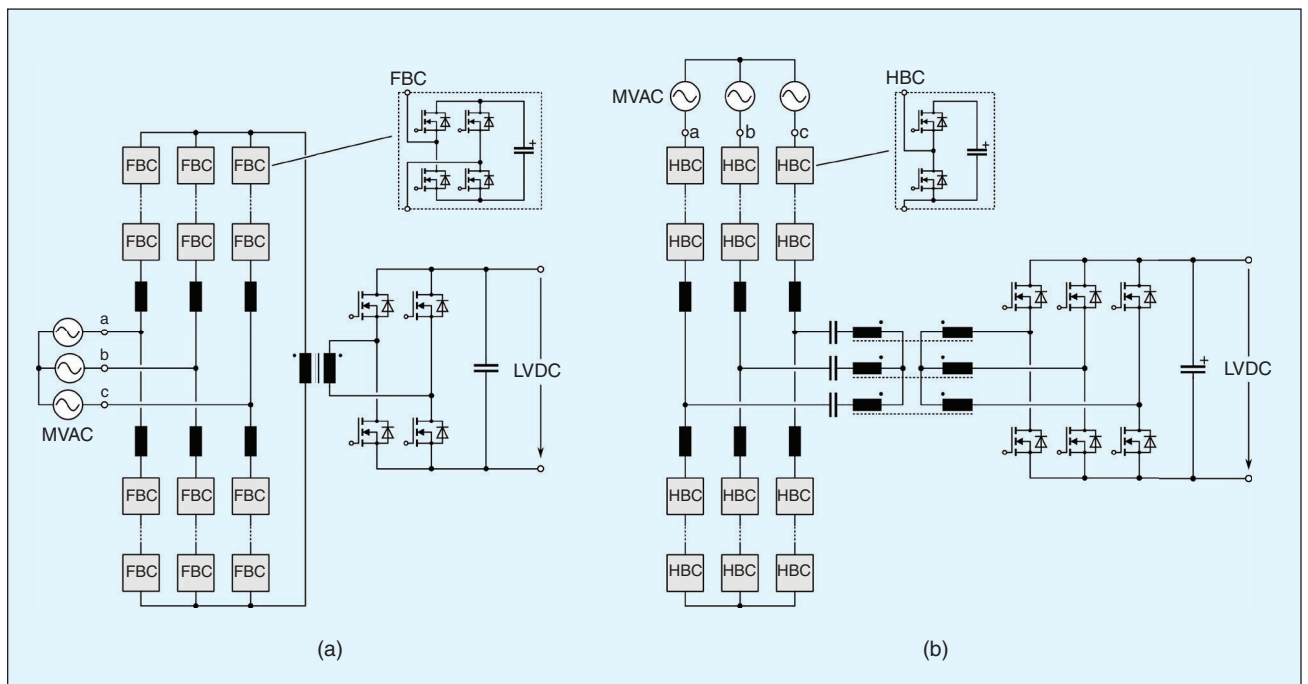


FIGURE 14 – Partly modular single-stage SSTs. (a) Direct three-phase to single-phase MMC matrix converter SST [48], [49] and (b) phase-modular SST using Q2L-operated MMC bridge legs and a single three-phase MFT [50].

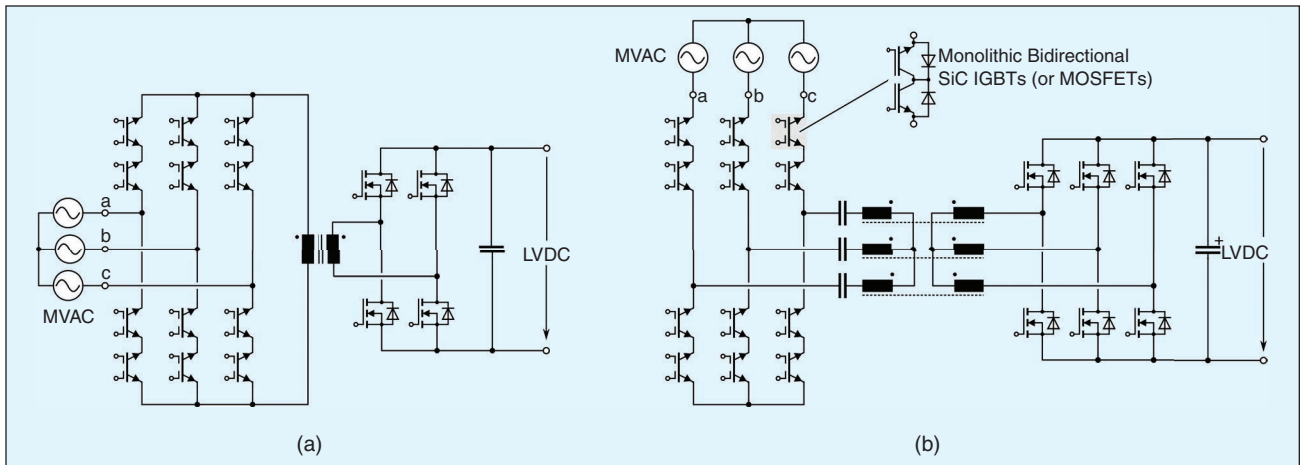


FIGURE 15 – Future monolithic single-stage SSTs, corresponding to the partly modular variants shown in Figure 14 but employing future high-voltage monolithic bidirectional power transistors. (a) Direct AC–DC matrix converter and (b) phase-modular single-stage isolated three-phase AC–DC converter.

with a phase-modular front end and a three-phase HF transformer [see Figure 15(b)] [51], [52], using either series-resonant or DAB-type operation. Note that both are monolithic (i.e., nonmodular) predecessors of the topologies discussed in the previous section and shown in Figure 14.

There is active research toward high-voltage (HV) monolithic bidirectional power transistors, e.g., 15-kV monolithic bidirectional SiC IGBTs [55]. Connecting, for example, two such devices in series per switch position and assuming a blocking voltage utilization of 75% would allow interfacing an MVAC mains with a line-to-line RMS voltage of up to 16 kV [the topology from Figure 15(a)] or 27 kV [the topology from Figure 15(b)]—with three

devices in series, today's often-used 35-kV MVAC voltage level is reachable for the phase-modular variant. Note that such HV monolithic bidirectional power transistors would also be highly suitable for the realization of the PSSs of the topologies discussed in the section “Quasi-Single-Stage Partly Modular SSTs.”

Finally, a series-resonant topology variant of Figure 15(b) employs a fully passive secondary-side diode rectifier and achieves PFC and sinusoidal mains currents with an extremely simple open-loop operation [56]. The LVDC output voltage is proportional to the MVAC input voltage, which might be tolerable for the input of the downstream converter stages if considered in the overall data center power distribution architecture accordingly.

with power ratings in the several-megawatts range are available [23]. Figure 16(b) shows an alternative using an LFT and a 12-pulse rectifier with an AF to achieve fully sinusoidal grid currents and reverse power flow (limited by the AF power rating of about 25%). Such systems are highly robust and have been employed for DC traction grid interties [57], electrolyzers, arc furnaces, etc. [58], and they can achieve very high efficiencies [3].

Similarly, the “Panama power supply” architecture [59], [60], widely employed in China, uses a multiwinding phase-shifting LFT to feed multiple secondary-side modules, each consisting of a three-phase diode rectifier and a regulation stage for the LVDC output. All LVDC outputs are connected to a common bus, and the phase-shifting transformer together with the high number of modules ensure low harmonic distortion of the MVAC input current even without an AF—similar approaches have been used for high-power MV drives.

In general, such LFT-based MVAC–LVDC interfaces are mature, and the transition to LVDC distribution would be possible immediately. Higher voltage levels than 800 VDC would be feasible, too, like the 1,500 VDC (typically ± 750 V, i.e., still within the LV regulatory framework) that is standard in PV systems today [10].

Finally, starting from LVAC distribution, the opposite path of *increasing* the distribution frequency,

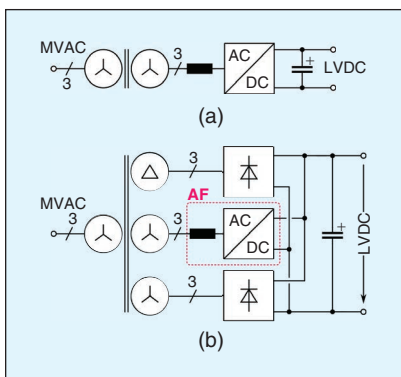


FIGURE 16 – State-of-the-art high-power MVAC–MVDC conversion based on combining a robust LFT with LV-side power electronics. (a) MVAC–LVAC LFT and downstream LV AC–DC converter. (b) LFT with 12-pulse rectifier and AF, enabling sinusoidal mains currents and reverse power flow of up to 25% [3], [57].

LFT-Based Short-Term Solutions

Whereas interest from both industry and academia in next-generation SSTs for data center applications is surging, data center concepts also foresee LVDC distribution [see Figure 2(b)] as a more near-term solution [6]. With the MVAC–LVDC conversion not being close to the servers, the requirements regarding compactness are less relevant, and it is worthwhile to briefly consider alternatives to SSTs.

Figure 16(a) shows the standard solution—an LFT and a downstream high-power LV AC–DC converter—for today's high-power PV plants [10], where containerized solutions

e.g., to 500 Hz, was proposed in the 1990s for sections of industrial grids [61]. Central frequency converters, e.g., SSTs [62] or, in the future, direct three-phase-to-three-phase matrix converters using only nine HV monolithic bidirectional power transistors, would be combined with passive and compact (because of the higher distribution frequency) LFT-based 12-pulse MVAC–LVDC conversion units next to the server racks, while centralized AFs would ensure grid-friendly behavior of the entire data center.

Conclusion and Outlook

Given the highly dynamic developments in IT and AI, forecasting of future data center energy consumption is a tricky business, as illustrated by Figure 17. A recent summary [63] identified widely differing projections for U.S. data center electricity consumption in 2030, i.e., in about four years, spanning a factor-of-five range between 200 and 1,050 TWh/a, with a concentration to around 500 TWh/a—corresponding to about 10% of the projected U.S. electricity consumption [64]. Thus, the future demand for data center power supply systems, and hence SSTs, is accordingly uncertain, and technology leaps in other fields could have disruptive impacts: for example, new optical computing approaches promise a hundredfold lower energy consumption per operation [65].

Nevertheless, immense investments are currently being poured into AI-related infrastructure, driven by the mentality that “misspending a couple of hundred billion” (M. Zuckerberg) is the lesser evil than missing an opportunity [66]. Infineon estimates that the global market for power semiconductors for SSTs reaches up to US\$1 billion [67]. Data center electrification thus remains a highly dynamic and highly interesting topic in power electronics and related disciplines, already leading to a shortage of skilled workers for AI data centers [68]. In addition to the power distribution architecture, there are other fascinating research questions emerging from the very fast and synchronized load

transients characteristic of AI model training runs [6], [69], which likely require highly dynamic power pulsation buffering close to the servers [70], possibly with new storage technologies [71] and/or buffer concepts leveraging a coordination of the AI workload and the power supply [72].

In this article, we have presented an overview of next-generation (quasi) single-stage MVAC–LVDC SST topologies that have the potential to break through the power density barrier characterizing today’s state-of-the-art SST demonstrator systems and to achieve a power density of 1 MW/m³ at an efficiency of 99%. This would facilitate low-loss MVAC distribution *within* the data center and up to the server racks. Further research should comparatively evaluate these (and possibly other) solutions, addressing topics like two-stage versus single-stage or quasi-single-stage topologies, new HV monolithic bidirectional power transistors, lifecycle environmental impacts (carbon footprint, resource usage, recycling, etc.) [73], protection and fault isolation inside of an SST as well as the implementation of redundancy concepts, and the relevance/applicability of today’s regulations (e.g., regarding isolation coordination requirements of MFTs in an SST, etc.). The research/solution space for future data center power distribution architectures follows the well-known X-Concepts in power electronics [74] (see Figure 18), e.g., hybridization (combination of LFTs and/or LF diode rectifiers with HF isolated DC–DC converters), modularization (monolithic versus phase/partially/fully modular SST topologies), and decentralization (centralized MVAC–LVDC conversion with LVDC distribution versions

distributed MVAC–LVDC SSTs close to the server racks).

A further and especially interesting data center distribution architecture combines the vectors of moving from AC to DC *and* from lower to higher voltages: MVDC distribution with central MVAC–MVDC rectification (e.g., based on MMCs as known from HVDC interties) and local MVDC–LVDC SSTs placed near the servers (see Figure 19) [77]. Even though this implies DC voltages outside of today’s LV regulatory framework, similar efforts are currently underway for PV applications, with a recent pilot project demonstrating inverters for a 3-kV DC string architecture [78]. Future holistic research should thus not only address MVAC–LVDC SSTs but also MVDC–LVDC SSTs,

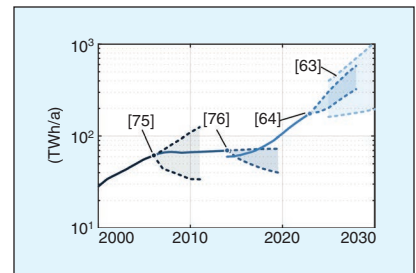


FIGURE 17 – U.S. data center electricity use (historical data and projections). Estimates for 2030 differ widely, but many fall into the 300–400 TWh/a range [63].

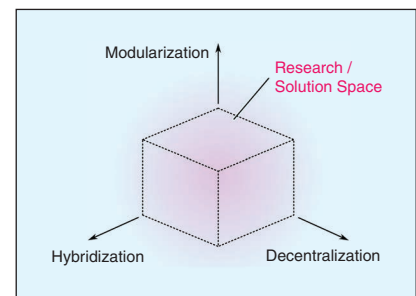


FIGURE 18 – Conceptual representation of the main degrees of freedom spanning the research/solution space for future data center power distribution architectures.

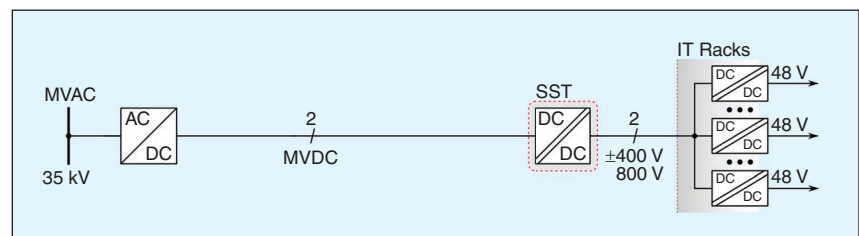


FIGURE 19 – Future MVDC power distribution architecture employing central MV rectification and localized high-power MVDC–LVDC SSTs.

MVDC and LVDC protection concepts (leveraging current-limiting capabilities of SSTs and/or possibly based on MV or LV SSCBs), and MVDC grid stability aspects together with local power/energy buffering concepts; standardization should then follow based on the results.

Finally, remember that SSTs are also applicable in fascinating applications other than data centers [14], e.g., megawatt charging stations for EVs and trucks, utility-scale PV installations, hydrogen production with water electrolysis, powering of high-power mining trucks, and shore-to-ship power supplies, opening a wide research area.

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